

$$112.13.i4. \rightarrow$$

$$\begin{aligned} & \frac{1}{6} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} + \frac{1}{16\pi^2} \left(\frac{1}{1296} \frac{1}{m_e^2} (-972 s_\gamma^2 c_\gamma^2 \bar{y}_d^{r i3} \bar{y}_d^{i2p} y_d^{r i4} y_d^{i1p} + s_\gamma^2 (-81 \bar{y}_d^{pr} \bar{y}_d^{i2i3} \right. \\ & \quad (y_d^{p i4} y_d^{i1r} (1 + s_\gamma^2) - 8 c_\gamma^2 y_d^{pr} y_d^{i1i4}) + 27 \bar{y}_d^{i2i3} (2 y_d^{i1i4} (g_1^2 - 3 (g_2^2 + 16 g_3^2)) + \\ & \quad y_d^{p i4} \bar{y}_d^{pr} y_u^{i1r} (-1 + c_\gamma^2)) - \bar{y}_d^{p i3} (81 \bar{y}_d^{i2r} y_d^{pr} y_d^{i1i4} (1 + s_\gamma^2) + \\ & \quad 27 \bar{y}_d^{i2r} (-y_d^{i1i4} y_u^{pr} (-1 + c_\gamma^2) + 12 c_\gamma^2 y_d^{p i4} y_u^{i1r}) + 2 g_1^2 y_d^{p i4} \delta_{i1i2}) + \\ & \quad 2 g_1^2 (-5 s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} + 13 c_\gamma^2 \bar{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4} - \\ & \quad \frac{1}{12} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_2 c_\gamma + s_\gamma c_2 c_\gamma) L F_{1,0}[m_d^p] - \\ & \quad \frac{2}{243} \sum_p g_1^4 L F_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{8}{324} \sum_p g_1^4 L F_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{8}{1215} \sum_p g_1^4 L F_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{pr} \bar{y}_d^{i2i3} y_d^{pr} y_d^{i1i4} (-s_2 c_\gamma + s_\gamma^3) L F_{1,0}[m_d^r] - \\ & \quad \frac{1}{12} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_2 c_\gamma + s_\gamma c_2 c_\gamma) L F_{1,0}[m_e^p] - \\ & \quad \frac{81}{243} \sum_p g_1^4 L F_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{108} \sum_p g_1^4 L F_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{8}{405} \sum_p g_1^4 L F_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{6} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \bar{y}_e^{pr} y_e^{pr} (-s_2 c_\gamma + s_\gamma^3) L F_{1,0}[m_e^r] + \\ & \quad \frac{1}{12} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 \bar{y}_e^{pr} y_e^{pr} (s_2 c_\gamma + s_\gamma^3) + \sum_p g_1^2 (2 s_2 c_\gamma + s_\gamma c_2 c_\gamma)) L F_{1,0}[m_e^p] - \\ & \quad \frac{1}{81} \sum_p g_1^4 L F_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{216} \sum_p g_1^4 L F_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{405}{405} \sum_p g_1^4 L F_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{1}{12} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \\ & \quad (6 \bar{y}_d^{pr} y_d^{pr} (-s_2 c_\gamma + s_\gamma^3) + 6 c_\gamma \bar{y}_u^{pr} y_u^{pr} (s_2 c_\gamma + s_\gamma c_2 c_\gamma) + \sum_p g_1^2 (2 s_2 c_\gamma + s_\gamma c_2 c_\gamma)) \\ & \quad L F_{1,0}[m_e^p] - \frac{1}{243} \sum_p g_1^4 L F_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{648} \sum_p g_1^4 L F_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{4}{1215} \sum_p g_1^4 L F_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{6} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} (2 s_2 c_\gamma + s_\gamma c_2 c_\gamma) L F_{1,0}[m_e^p] - \\ & \quad \frac{243}{243} \sum_p g_1^4 L F_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{81} \sum_p g_1^4 L F_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{32}{1215} \sum_p g_1^4 L F_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{2} s_\gamma c_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \bar{y}_u^{pr} y_u^{pr} (s_2 c_\gamma + s_\gamma c_\gamma) L F_{1,0}[m_e^r] - \frac{1}{24} s_\gamma \frac{1}{m_e^4} \bar{y}_d^{i2i3} y_d^{i1i4} \\ & \quad (3 s_4 c_\gamma (g_1^2 + g_2^2) + s_\gamma (g_1^2 (-1 + 3 c_2 c_\gamma^2) + 3 g_2^2 (-1 + c_2 c_\gamma^2))) L F_{1,0}[m_e] + \\ & \quad \frac{1}{24} \frac{1}{m_e^2} s_\gamma^2 (3 s_\gamma^2 \bar{y}_d^{pr} \bar{y}_d^{i2i3} y_d^{p i4} y_d^{i1r} + 3 \bar{y}_d^{p i3} y_d^{i1i4} (s_\gamma^2 \bar{y}_d^{i2r} y_d^{pr} - c_\gamma^2 \bar{y}_u^{i2r} y_u^{pr}) - \\ & \quad \bar{y}_d^{i2i3} (2 y_d^{i1i4} (g_1^2 + 3 g_2^2) + 3 c_\gamma^2 y_d^{p i4} \bar{y}_u^{pr} y_u^{i1r})) L F_{1,1}[m_e] + \frac{1}{216} \\ & \quad (s_\gamma^2 (-108 c_\gamma^2 \bar{y}_d^{r i3} \bar{y}_d^{i2p} y_d^{r i4} y_d^{i1p} + 9 \bar{y}_d^{i2i3} (y_d^{i1i4} (g_1^2 + 3 g_2^2) + 2 c_\gamma^2 y_d^{p i4} \bar{y}_u^{pr} y_u^{i1r}) + \\ & \quad 2 \bar{y}_d^{p i3} (9 c_\gamma^2 \bar{y}_u^{i2r} (y_d^{i1i4} y_u^{pr} - 6 y_d^{p i4} y_u^{i1r}) - 2 g_1^2 y_d^{p i4} \delta_{i1i2})) - \\ & \quad 8 g_1^2 (s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - 2 c_\gamma^2 \bar{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4}) L F_{1,2}[m_e] + \\ & \quad \frac{1}{36} (-9 s_\gamma^4 \bar{y}_d^{r i3} \bar{y}_d^{i2p} y_d^{r i4} y_d^{i1p} + s_\gamma^2 \bar{y}_d^{p i3} y_d^{p i4} (9 c_\gamma^2 \bar{y}_u^{i2r} y_u^{i1r} + g_1^2 \delta_{i1i2}) + \\ & \quad g_1^2 (s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - c_\gamma^2 \bar{y}_u^{i2p} y_u^{i1p}) \delta_{i3i4}) L F_{2,1}[m_e] - \\ & \quad \frac{1}{324} g_1^2 (9 s_\gamma^2 \bar{y}_d^{p i3} y_d^{p i4} \delta_{i1i2} + (9 s_\gamma^2 \bar{y}_d^{i2p} y_d^{i1p} - 9 c_\gamma^2 \bar{y}_u^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \\ & \quad L F_{3,0}[m_e] + \frac{5}{216} g_1^4 L F_{4,-1}[m_e] \delta_{i1i2} \delta_{i3i4} - \frac{4}{405} g_1^4 L F_{5,-2}[m_e] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{81}{81} g_1^4 L F_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{54} g_1^4 L F_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{8}{405} g_1^4 L F_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} L F_{1,1,0}[m_1, m_d^{i3}] - \\ & \quad \frac{1}{108} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} L F_{2,1,-1}[m_1, m_d^{i3}] - \frac{1}{972} g_1^4 L F_{2,1,0}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{972} g_1^4 L F_{2,2,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{486} g_1^4 L F_{3,1,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{972} g_1^4 L F_{4,1,-2}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{54} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} L F_{1,1,0}[m_1, m_d^{i4}] - \\ & \quad \frac{1}{108} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} L F_{2,1,-1}[m_1, m_d^{i4}] - \frac{1}{972} g_1^4 L F_{2,1,0}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{972} g_1^4 L F_{2,2,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{486} g_1^4 L F_{3,1,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{972} g_1^4 L F_{4,1,-2}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{216} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \bar{y}_d^{i2i3} y_d^{i1i4} L F_{$$